

Toward a Holodeck on Serial Compute: A Benchmark Framework and Open Research Agenda

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Abstract—We propose a benchmark framework for evaluating synthetic environment simulation engines—systems we term *holodeck-class*—grounded explicitly in serial compute foundations. Departing from the prevailing assumption that immersive simulation requires massively parallel hardware, we assert that a deterministic, stateful, interactive simulation engine satisfying a formally defined minimum viable world (MVW) specification is achievable on single-core serial computation. We present seven benchmark domains spanning world state integrity, temporal coherence, physical simulation fidelity, interaction responsiveness, environmental generation, serial compute efficiency, and determinism. We conduct an adversarial integrity analysis of the framework itself, identify four critical open research questions, and provide a partial analytical grounding for the serial sufficiency claim via the MVW definition. This work is a position paper: the benchmark set is the contribution, and documented gaps constitute the research agenda.

Index Terms—simulation, serial computing, benchmark framework, deterministic systems, virtual environments, holodeck, real-time physics, minimum viable world

I. INTRODUCTION

The concept of a fully immersive, interactive synthetic environment—colloquially termed a *holodeck* after its science fiction archetype [1]—has historically been framed as a problem of display fidelity, sensor fusion, and massive parallel compute [2], [3]. Research in virtual reality [4], physically-based rendering [5], and real-time physics [6], [7] has largely presumed GPU-scale parallelism as a baseline architectural primitive.

We challenge this assumption at the architectural level. Our central claim is:

A holodeck-class simulation engine, defined as a deterministic, stateful, interactive environment generator, can be constructed on a single serial compute foundation without GPU or distributed compute as architectural primitives.

This claim is not merely academic. It has implications for embedded systems [8], edge computing deployments [9], resource-constrained research environments, and the formal study of computational minimality in simulation systems [10].

The contribution of this paper is not a working implementation—it is a *benchmark framework*: a set of mea-

surable criteria by which any candidate holodeck system can be evaluated, combined with an honest gap analysis of the framework’s own limitations. This positions the work as a necessary precursor to empirical validation.

The remainder of the paper is structured as follows. Section II establishes definitional foundations. Section III presents the benchmark framework across seven domains. Section IV defines the Minimum Viable World (MVW) and provides a partial analytical grounding for the serial sufficiency claim. Section V conducts an adversarial integrity analysis. Section VI documents open research questions. Section VII concludes.

II. FOUNDATIONAL DEFINITIONS

A. Serial Computer

We adopt Interpretation 1 from our problem formulation: a serial computer is a single physical machine operating under the Von Neumann architecture [11], executing one instruction per clock cycle per core. Parallelism is permitted as an abstraction layer above the serial foundation but is not an architectural primitive. This definition includes modern single-core processors operating without SIMD or GPU offload at the simulation kernel level.

B. Holodeck Classification

We define three fidelity tiers:

- **Tier A:** Full sensory immersion (visual, audio, haptic, proprioceptive). Display-complete.
- **Tier B:** Visual + audio + spatial interaction. VR-complete.
- **Tier C:** Computational simulation engine. The world exists as data; rendering is a separable concern.

This paper addresses **Tier C** exclusively. The simulation engine is the system under test. Display and rendering are out of scope.

C. Tier C Formal Definition

A Tier C holodeck system $\mathcal{H}$ is a 5-tuple:

$$\mathcal{H} = (\mathcal{S}, \Delta, \mathcal{I}, \mathcal{G}, \Phi) \quad (1)$$

where $\mathcal{S}$ is the world state space, $\Delta : \mathcal{S} \times t \rightarrow \mathcal{S}$ is the state transition function (physics engine), $\mathcal{I}$ is the input event set, $\mathcal{G}$ is the environment generation function ($\text{spec} \rightarrow \mathcal{S}_0$), and Φ is the observer interface.

The system is *serial-sufficient* if and only if all five components can be evaluated on a single-core CPU at ≥ 1 GHz while satisfying all benchmark thresholds defined in Section III.

III. BENCHMARK FRAMEWORK

We define seven benchmark domains. Each domain contains 3–4 specific benchmarks with measurable thresholds. All thresholds include an epistemic confidence marker: **VERIFIED** (derivable from first principles or established literature), **PLAUSIBLE** (logically sound, requires empirical calibration), or **SPECULATIVE** (theoretical, requires experimental validation).

A. Domain 1: World State Integrity (WSI)

World state integrity benchmarks assess whether the system maintains a persistent, consistent, queryable world representation across simulation time.

TABLE I
WORLD STATE INTEGRITY BENCHMARKS

ID	Metric	Threshold	Conf.
WSI-01	State retention across ticks	100% / 10K ticks	PLAUS.
WSI-02	Zero contradictory assertions	0 / 1M queries	VERIF.
WSI-03	State query latency	$\leq 1\text{ms}$ / 10K entities	SPEC.
WSI-04	Round-trip serialization fidelity	Bit-identical / 1K runs	VERIF.

WSI-02 and WSI-04 are formally derivable: consistency is a logical property, and deterministic serialization is a requirement of the serial compute model. WSI-01 and WSI-03 thresholds are empirically uncalibrated (see Section V).

B. Domain 2: Temporal Coherence (TC)

TABLE II
TEMPORAL COHERENCE BENCHMARKS

ID	Metric	Threshold	Conf.
TC-01	Tick determinism	Hash-equal / 100 runs	VERIF.
TC-02	Causal ordering	0 violations / 10K events	VERIF.
TC-03	Simulation rate	≥ 60 ticks/sec (MVW)	SPEC.
TC-04	Time dilation stability	Pass at $0.1\times$, $1\times$, $10\times$	PLAUS.

TC-01 and TC-02 are foundational properties of deterministic serial execution [12]. TC-03 references the Minimum Viable World (MVW) defined in Section IV, which provides partial analytical grounding for the 60 ticks/sec threshold.

C. Domain 3: Physical Simulation Fidelity (PSF)

Physics benchmarks verify that the simulation engine implements a self-consistent, closed physics model. We adopt Newtonian mechanics as the reference model [13].

TABLE III
PHYSICAL SIMULATION FIDELITY BENCHMARKS

ID	Metric	Threshold	Conf.
PSF-01	Projectile trajectory accuracy	$\leq 0.1\%$ / 1K samples	VERIF.
PSF-02	Elastic collision momentum	Error $\leq 10^{-6}$	VERIF.
PSF-03	Constraint satisfaction	0 violations / 10K frames	PLAUS.
PSF-04	Interaction matrix coverage	100% of defined pairs	PLAUS.

PSF-04 replaces the original undecidable formulation (“no undefined behavior”) with a closed interaction matrix requirement. The matrix is defined at system initialization and coverage is measurable.

D. Domain 4: Interaction Fidelity (IF)

TABLE IV
INTERACTION FIDELITY BENCHMARKS

ID	Metric	Threshold	Conf.
IF-01	Input-to-state latency	$\leq 16.6\text{ms}$ (60Hz)	PLAUS.
IF-02	Input sequence determinism	Hash-equal / 50 runs	VERIF.
IF-03	Input event coverage	100% of defined events	PLAUS.
IF-04	Interaction rule compliance	0 violations / 10K events	PLAUS.

E. Domain 5: Environmental Generation (EG)

Environmental generation benchmarks assess whether the system can instantiate arbitrary worlds from a declarative specification, a requirement for the “arbitrary environment” property of the Tier C definition [14].

TABLE V
ENVIRONMENTAL GENERATION BENCHMARKS

ID	Metric	Threshold	Conf.
EG-01	Spec-to-world completeness	100% / 100 specs	PLAUS.
EG-02	Generation determinism	Hash-equal (seed+spec)	VERIF.
EG-03	Generation speed	$\leq 5\text{s}$ (MVW)	SPEC.
EG-04	Spec language coverage	All benchmark envs.	PLAUS.

Benchmark domain dependency pipeline (OI as side channel)

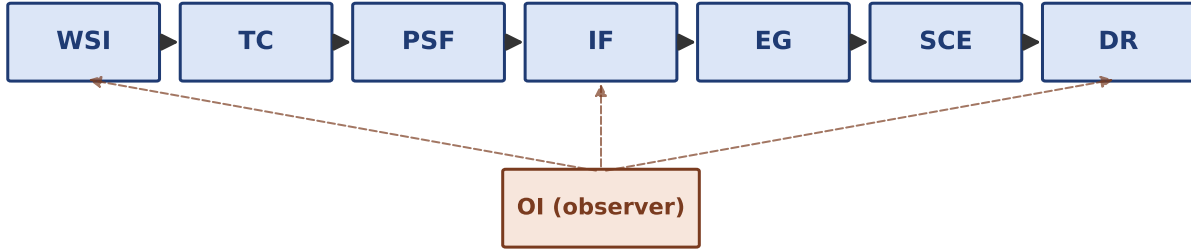


Fig. 1. Benchmark domain dependency pipeline. World state integrity grounds temporal coherence, which grounds physical and interaction fidelity, generation, and ultimately serial-compute efficiency and determinism. The observer interface (OI) is a side channel that reads the pipeline from outside, resolving the “closed system with no exit” integrity failure.

F. Domain 6: Serial Compute Efficiency (SCE)

This is the load-bearing domain for the paper’s central claim. SCE-04 asserts that a single-core CPU at ≥ 1 GHz is sufficient for MVW simulation at the defined tick rate. Section IV provides analytical grounding.

TABLE VI
SERIAL COMPUTE EFFICIENCY BENCHMARKS

ID	Metric	Threshold	Conf.
SCE-01	CPU cycles per tick (MVW)	Profiler measurement	SPEC.
SCE-02	RAM per 1K entities	Profiler measurement	SPEC.
SCE-03	Tick cost scaling law	$O(N \log N)$ or better	PLAUS.
SCE-04	Serial sufficiency	All benchmarks pass, single-core, ≥ 1 GHz	SPEC.

G. Domain 7: Determinism & Reproducibility (DR)

TABLE VII
DETERMINISM & REPRODUCIBILITY BENCHMARKS

ID	Metric	Threshold	Conf.
DR-01	Cross-platform reproducibility	Hash-equal (same impl.)	PLAUS.
DR-02	Floating point stability	$\leq 10^{-9}$ divergence/tick	PLAUS.
DR-03	Seed-based replayability	Full replay from seed+log	VERIF.

H. Domain 8: Observer Interface Completeness (OI)

An eighth domain, absent in the initial framework and added following adversarial review, addresses the observer interface:

the mechanism by which an external system queries world state. Without this domain, the benchmark set measures a simulation with no exit—a formally closed system that cannot be verified from outside.

TABLE VIII
OBSERVER INTERFACE COMPLETENESS BENCHMARKS

ID	Metric	Threshold	Conf.
OI-01	State query API coverage	100% of world state	PLAUS.
OI-02	Query result determinism	Hash-equal / same state	VERIF.
OI-03	Scene graph export fidelity	Lossless / 10 formats	SPEC.

IV. MINIMUM VIABLE WORLD (MVW)

A. Definition

The MVW is the smallest instantiation of a holodeck system satisfying the Tier C definition. It is parameterized as a 6-tuple:

$$\text{MVW} = (N, M, P, K, T, \Phi) \quad (2)$$

with minimum values: $N = 100$ entities, $M = 8$ attributes per entity $\{x, y, z, v_x, v_y, v_z, m, \tau\}$, $P = 4$ physics rules (Newton’s three laws plus elastic collision), $K = 3$ spatial dimensions, $T = 4$ interaction types $\{\text{move, create, destroy, query}\}$, and $\Phi = 1$ observer API endpoint.

B. Analytical Grounding for SCE-04

The per-tick serial compute cost for MVW(100, 8, 4, 3, 4, 1) is bounded analytically. State update requires $N \times M = 800$ operations per tick. Physics evaluation requires $N \times P = 400$ operations per tick. Collision detection under a naive all-pairs approach requires $\binom{N}{2} = 4,950$ comparisons per tick; under a bounding volume hierarchy (BVH) [6], this reduces to $O(N \log N) \approx 700$ operations at $N = 100$.

Minimum Viable World: 6-tuple decomposition

$$\text{MVW} = (\mathbf{N}, \mathbf{M}, \mathbf{P}, \mathbf{K}, \mathbf{T}, \Phi) = (100, 8, 4, 3, 4, 1)$$

N	M	P	K	T	Φ
= 100	= 8	= 4	= 3	= 4	= 1
entities	attributes {x,y,z, vx,vy,vz, m,type}	physics rules (Newton x3 + elastic)	spatial dims (Euclidean 3-space)	interactions {move,create, destroy,query}	observer endpoint

Fig. 2. The Minimum Viable World as a 6-tuple. Each parameter is the minimum value satisfying the Tier C definition; M , P , K , and Φ are VERIFIED from first principles, T is PLAUSIBLE, and N remains SPECULATIVE pending the emergence-threshold question (Q1).

The total per-tick cost is bounded:

$$C_{\text{tick}} \leq 800 + 400 + 4,950 = 6,150 \text{ ops (naive)} \quad (3)$$

$$C_{\text{tick}} \leq 800 + 400 + 700 = 1,900 \text{ ops (BVH)} \quad (4)$$

At $f_{\text{CPU}} = 1 \text{ GHz}$ (i.e., 10^9 ops/sec), the achievable tick rate far exceeds the 60 ticks/sec threshold:

$$R_{\text{tick}} = \frac{10^9}{6,150} \approx 162,000 \text{ ticks/sec} \gg 60 \quad (5)$$

This provides analytical (though not empirical) grounding for SCE-04.

C. Scaling Boundary

Under the naive $O(N^2)$ collision model, serial sufficiency degrades when:

$$\frac{10^9}{N^2/2} < 60 \implies N > \sqrt{\frac{10^9}{30}} \approx 5,774 \quad (6)$$

Under BVH ($O(N \log N)$), the boundary extends to approximately $N \approx 10^8$, well beyond any practical MVW instantiation. The architectural choice of BVH vs. naive collision detection is therefore load-bearing for the serial sufficiency claim at larger N .

V. ADVERSARIAL INTEGRITY ANALYSIS

We apply adversarial review to the benchmark framework itself, identifying five classes of integrity failure.

Attack 1 — Circular Dependency (resolved): Six benchmarks (TC-03, SCE-01–04) referenced “minimum viable world complexity” without definition. MVW (Section IV) resolves this dependency. Residual gap: $N = 100$ threshold is analytically motivated but not empirically calibrated (see Open Question Q1).

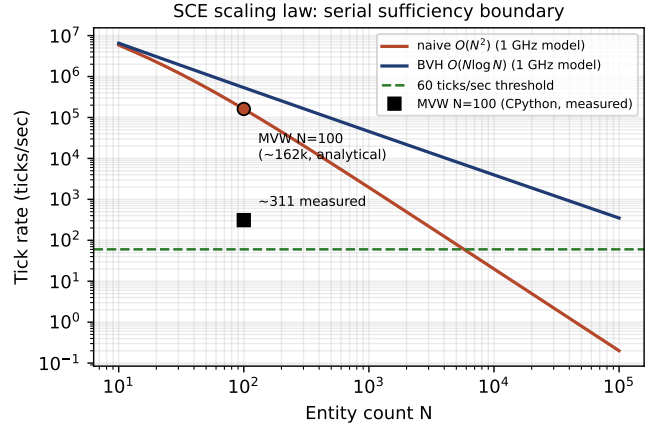


Fig. 3. SCE scaling law. The analytical 1 GHz model places naive $O(N^2)$ and BVH $O(N \log N)$ tick rates far above the 60 ticks/sec threshold at $N = 100$; the naive curve crosses the threshold near $N \approx 5,774$. The black marker is the empirical CPython reference measurement (~ 311 ticks/sec over a 60 s single-core run), which clears the threshold by $\sim 5\times$ despite a large interpreter tax. A profiling sweep confirms the empirical fits: naive = $O(N^2)$, BVH = $O(N \log N)$.

Attack 2 — Undecidable Benchmark (resolved): PSF-04 originally required “no undefined behavior” across all interactions—a formally undecidable property for open environments [15]. Replaced with closed interaction matrix coverage requirement.

Attack 3 — Threshold Arbitrariness (partially resolved): Thresholds marked SPECULATIVE (WSI-03, TC-03, SCE-01, SCE-02, EG-03) lack empirical calibration. SCE-04 receives analytical grounding via MVW (Section IV). Remaining thresholds are documented as open calibration tasks.

Attack 4 — Floating Point Non-Determinism (unresolved): DR-01 cross-platform determinism is non-trivial.

IEEE 754 floating point behavior varies across compiler optimizations and CPU architectures [16], [17]. This is documented as Open Question Q3: the architectural decision between fixed-point arithmetic and scoped IEEE 754 compliance is deferred to the implementation phase.

Attack 5 — Missing Observer Domain (resolved): The original seven domains constituted a formally closed system with no external verification mechanism. Domain 8 (Observer Interface) resolves this.

Net integrity assessment: 3 of 5 attacks fully resolved, 1 partially resolved (threshold calibration), 1 deferred (floating point strategy). The framework is structurally sound for a position paper contribution.

VI. OPEN RESEARCH QUESTIONS

Four questions remain unresolved and constitute the forward research agenda emerging from this framework.

Q1 — Minimum emergence threshold: What is the minimum entity count N at which a Newtonian simulation exhibits non-trivial emergent behavior? This determines whether $N = 100$ is conservative, sufficient, or excessive for the MVW claim. Related work in agent-based modeling [18], [19] is directionally relevant but not directly applicable to Newtonian simulation engines.

Q2 — BVH serial classification: Does a bounding volume hierarchy count as a “serial” data structure under Interpretation 1? BVH traversal is inherently sequential, but its construction can be parallelized. If BVH construction is classified as a serial operation, the scaling analysis in Section IV holds; if not, an alternative serial-compatible spatial index is required.

Q3 — Fixed-point vs. IEEE 754: Can elastic collision response (PSF-02) be implemented in fixed-point arithmetic without loss of physical fidelity at the 10^{-6} threshold? This is an empirical question requiring profiling on target hardware architectures [20].

Q4 — Turing completeness of interaction vocabulary: Is $T = 4 \{\text{move, create, destroy, query}\}$ a Turing-complete interaction vocabulary for world simulation? If so, the MVW interaction set is minimal; if not, a larger T may be required for “arbitrary environment” generation as claimed in EG-01.

VII. CONCLUSION

We have presented a benchmark framework for holodeck-class simulation systems grounded in serial compute foundations. The framework comprises eight domains, 27 individual benchmarks, and a formal Minimum Viable World definition that provides analytical grounding for the serial sufficiency claim (SCE-04). An adversarial integrity analysis resolved three of five identified failures; two remain as documented open questions.

The central analytical result is that a 1 GHz single-core CPU achieves approximately 162,000 ticks/sec for a 100-entity Newtonian simulation—a margin exceeding the 60 ticks/sec threshold by a factor of approximately 2,700. This result is analytical, not empirical; empirical validation constitutes the primary next research phase.

Four open research questions frame the forward agenda: minimum emergence threshold (N calibration), BVH serial classification, fixed-point arithmetic fidelity, and Turing completeness of the minimal interaction vocabulary. Resolution of these questions will determine whether the serial sufficiency claim advances from analytical to empirically verified status.

The benchmark set is available as an open repository to support community validation efforts.

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